IEEE Open Journal of
Power Electronics

Received XX Month, XXXX; revised XX Month, XXXX; accepted XX Month, XXXX; Date of publication XX Month, XXXX; date of current version XX Month, XXXX.

Digital Object Identifier 10.1109/OJPEL.2022.0817315

Finite control set model predictive control for current ripple reduction on BLDCs

AXEL CORONADO-ANDRADE*, AND LUIS ALVAREZ-ICAZA†, MEMBER, IEEE

[1]Facultad de Ingeniería, Universidad Nacional Autónoma de México, CDMX 04510,Mexico
[2]Instituto de Ingeniería, Universidad Nacional Autónoma de México, CDMX 04510, Mexico

CORRESPONDING AUTHOR: Luis Alvarez-Icaza (e-mail: alvar@pumas.iingen.unam.mx).

ABSTRACT Two new techniques are presented for finite control set model predictive control (FCS-MPC) of BLDC motors in high performance applications. The proposals are based on two design choices: a) a reduced set of admissible switching states, selected according to the electrical sector in which the motor is functioning, and b) the division of the sampling time into a fixed number of sub-intervals. This two choices, together with a proper formulation for the prediction of current values, allow to easily calculate the value of a predefined cost function at the end of the sampling time, for all possible combinations of admissible switchings during the sub-intervals, and to select the one that minimizes this cost function. The chosen combination is then applied using a standard voltage source inverter (VSI). These two techniques do not require the calculation of any switching time and keep the simplicity of the original FCS-MPC scheme. Simulation and experimental results, under similar conditions, show a superior performance with a reduction between 75% to 80% of the RMSE current tracking error, when compared with the original FCS-MPC.

INDEX TERMS BLCD motors, current ripple reduction, Finite control set model predictive control, voltage source inverter.

I. INTRODUCTION

IN recent years, Brushless Direct Current (BLDC) motors have gained significant popularity across various industrial and commercial applications, particularly in the automotive sector. This is largely due to their high efficiency and power density, and lower maintenance requirements compared to traditional motors. [1]–[3]. A key characteristic of BLDC motors, imposing challenges in the control of currents and speed, is their back electromotive force (BEMF) which, depending on the stator winding arrangement, can be trapezoidal or sinusoidal [4]. One of the most critical issues dealing with BLDC motors featuring trapezoidal back EMF is their torque ripple, which can degrade performance, shorten lifespan, and reduce precision in high performance or smooth torque control applications. Torque ripple originates not only from the EMF shape, it is also due to the cogging torque that depends on the motor internal structure, and the chosen control and commutation strategy [5].

In industrial applications, the most widely used control techniques are Six-Step Commutation and Field-Oriented Control (FOC). The Six-Step method is simple and efficient, however introduces significant torque ripple, especially at low speeds. For this reason, vector control techniques such as FOC are preferred [4]. Although FOC does not eliminate torque ripple due to the trapezoidal nature of the motor back EMF, it helps to reduce it. Other techniques explored in the literature include direct torque control (DTC) [6], [7], hysteresis controllers [8], [9], and model predictive control (MPC), all of which aim to further reduce torque ripple and improve dynamic response under various conditions.

MPC offers several advantages, such as ease of interpretation, online optimization, inclusion of constraints and nonlinearities, low current distortion, and minimal switching losses [10]–[13]. A specific implementation of MPC for BLDC motors is Finite Control Set Model Predictive Control (FCS-MPC) [14], which leverages the limited number of switching states in inverters by solving an optimization problem. Using a discrete model, it predicts the motor future behavior for each valid switching state and selects the one that minimizes a pre-defined cost function. One of the key advantages of FCS-MPC is that it directly applies control actions to the inverter without requiring a modulation

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stage, allowing for a fast dynamic response. However, it also presents challenges compared to FOC or DTC, such as variable switching frequency, and increased total harmonic distortion (THD) in current. When only one swichting combination, or voltage vector, can be applied per sampling period, achieving an accuracy similar to classical techniques requires high sampling frequencies; this becomes particularly problematic in low-inductance motors if the hardware for calculations cannot achieve sufficiently short sampling times.

To address these limitations, variations of FCS-MPC have been proposed in the literature, particularly for permanent magnet synchronous motors (PMSM) [15]–[22]. Instead of applying a single voltage vector per sampling time, these approaches combine multiple vectors, such as an active vector with a null vector, or two or more active vectors. For example, Li et al. [18] propose the use of multi-virtual-vectors combining one null vector and two active vectors, with activation times optimized using dead-beat control. Similarly, Zhang et al. [19] present a double-current-compensation model predictive current control (DCC-MPCC), combining active and null vectors, allowing up to four vectors per cycle, improving steady-state performance while reducing computational burden. Sheng et al. [20] introduce a two-vector FCS-MPC approach based on duty cycle ratio optimization, achieving improved dynamic response and lower torque ripple, though requiring real-time optimization. Moreover, Wang et al. [21] demonstrate that an improved double-vector strategy using three virtual vectors can reduce total harmonic distortion (THD) and lighten computational complexity. Zhao et al. [22] propose an enhanced dual-vector predictive control using optimal vector selection, which minimizes trigonometric computations while maintaining high control performance.

While these methods reduce the demand for extremely short sampling times, they also increase computational complexity by requiring additional calculations, for example, determining activation times to use with dead-beat control or other strategies. Furthermore, it is worth mentioning that most of the available research focuses on PMSM, while few works specifically address BLDC motors, which present additional challenges due to their trapezoidal back-EMF and inherent torque ripple.

This work presents two variations of FCS-MPC that only evaluate a finite set of vectors in the optimization of the cost function and does not require real-time computation of activation times. Additionally, these two approaches provide flexibility in adjusting the algorithm execution frequency according to hardware capability and the required tracking error.

To validate the effectiveness of the proposed techniques, they were compared with classical FCS-MPC in the current control of a relatively low-inductance BLDC motor with trapezoidal back EMF. Results indicate that the new techniques significantly reduce current tracking errors and can also reduce torque ripple when the current trajectories are properly selected according to the motor's back EMF shape. Moreover, this can be achieved without requiring extremely short sampling times, making the proposed methods more practical for high-performance implementations on hardware with limited processing capacity.

The rest of the paper includes the mathematical materials required for FCS-MPC in Section II, while in Section III describes the new two proposals for FCS-MPC introduced in this work. Sections IV and V contain, respectively, the simulation and experimental results using the two new proposals and the comparison of their performance with the clasical FCS-MPC [14]. Finally, section VI contains the final conclusions of the work and recommendations for future research.

II. MATHEMATICAL PRELIMINARIES FOR FCS-MPC

A. MATHEMATICAL MODEL OF THE BLDC MOTOR

The brushless DC motor under study has a permanent magnet rotor and three sets of windings on the inner surface of the stator connected in **Y** configuration. The BEMF of this motor has a trapezoidal shape as the magnetic fields are uniformly distributed around the rotor. The model of the BLDC motor in the $\alpha\beta$ frame can be expressed as follows [23]:

$$\frac{di_\alpha}{dt} = \frac{1}{L_s}(u_\alpha - e_\alpha - Ri_\alpha) \tag{1a}$$

$$\frac{di_\beta}{dt} = \frac{1}{L_s}(u_\beta - e_\beta - Ri_\beta) \tag{1b}$$

$$\frac{d\theta}{dt} = \omega \tag{1c}$$

$$\mathrm{J}\frac{d\omega}{dt} = T_e - \mathrm{d}\omega - \tau_L(t)\,, \tag{1d}$$

where $i_\alpha, i_\beta, u_\alpha, u_\beta, e_\alpha$ and e_β are the stator currents, voltages and the back electromotive force in the $\alpha\beta$ frame, $\omega \in \mathbb{R}$ the angular velocity, $T_e \in \mathbb{R}$ the electric torque, $\tau_\mathrm{L} \in \mathbb{R}$ the load torque, $\mathrm{L}_s \in \mathbb{R}$ the winding inductances, $\mathrm{R} \in \mathbb{R}$ the winding resistances, $\mathrm{J} \in \mathbb{R}$ the rotor inertia and $\mathrm{d} \in \mathbb{R}$ the viscous friction coefficient.

B. CURRENT PREDICTIONS AND COST FUNCTION IN FCS-MPC

FS-MPC is a control technique that takes advantage of the limited number of possible switching states in the inverters. The method uses a discrete time dynamic model to predict the future behavior of the motor currents for each admissible switching state. Then the switching action that minimizes a predefined cost function is selected and applied at the next sampling instant. As mentioned, the main advantage of this method is that each control action is precisely the inverter configuration, therefore it can applied directly to this device without requiring a modulation stage.

The discrete time dynamic model used in FCS-MPC is obtained discretizing equations (1a) and (1b) with sampling time T_s using Euler's approximation. Then, currents i_α and

i_β in the next sampling time can be predicted as

$$i_\alpha(k+1) = i_\alpha(k) + \frac{T_s}{L_s}(u_\alpha(k) - e_\alpha(k) - Ri_\alpha(k)) \quad (2a)$$

$$i_\beta(k+1) = i_\beta(k) + \frac{T_s}{L_s}(u_\beta(k) - e_\beta(k) - Ri_\beta(k)) \quad (2b)$$

The cost function for all FCS-MPC techniques used in this paper is

$$G = (i_{\alpha ref}(k+1) - i_\alpha(k+1))^2 + (i_{\beta ref}(k+1) - i_\beta(k+1))^2 \quad (3)$$

where $i_{\alpha ref}$ and $i_{\beta ref}$ are the reference currents in the $\alpha\beta$ frame.

C. SELECTION OF ADMISSIBLE VOLTAGE VECTORS

The number of switching states to be tested depends on the inverter used. In this work, a VSI is employed, which can generate eight different switching states. Table. I shows the state V_i associated with the position of the three switches S_a, S_b and S_c[1], the full binary state of the six switches, and the corresponding voltages u_α, u_β, with u_d the applied voltage.

TABLE I. VSI Voltage Vectors

V	S_a	S_b	S_c	State	u_α	u_β
V_0	0	0	0	0,1,0,1,0,1	0	0
V_1	1	0	0	1,0,0,1,0,1	$\frac{2u_d}{3}$	0
V_2	1	1	0	1,0,1,0,0,1	$\frac{u_d}{3}$	$\frac{u_d}{\sqrt{3}}$
V_3	0	1	0	0,1,1,0,0,1	$-\frac{u_d}{3}$	$\frac{u_d}{\sqrt{3}}$
V_4	0	1	1	0,1,1,0,1,0	$-\frac{2u_d}{3}$	0
V_5	0	0	1	0,1,0,1,1,0	$-\frac{u_d}{3}$	$-\frac{u_d}{\sqrt{3}}$
V_6	1	0	1	1,0,0,1,1,0,	$\frac{u_d}{3}$	$-\frac{u_d}{\sqrt{3}}$
V_7	1	1	1	1,0,1,0,1,0	0	0

Note that both V_0 and V_7 in Table. I produce a zero voltage vector, reducing the number of applied voltage vectors to 7. Only one of these admissible voltage vectors V_i can be injected at each sampling instant, making it essential to execute the algorithm as quickly as possible. Otherwise, the magnitude of the error will increase significantly. This constitutes one of the main limitations of this control technique, since the sampling time cannot be arbitrarily reduced due to hardware constraints during implementation.

D. CLASICAL FCS-MPC ALGORITHM

Algortihm 1 summarizes the choice of voltage vector for the normal FCS-MPC technique.

[1]Switches $S_{\bar{a}}$, $S_{\bar{b}}$ and $S_{\bar{c}}$ with complementary states are not included in Table. I.

Algorithm 1 FCS-MPC

Require: $i(k), u(k)$
  $G_{op} \leftarrow G_{max}$
  $j \leftarrow 1$
  for $j \leftarrow 1, 7$ **do**
    $u(j) \leftarrow [u_\alpha \quad u_\beta]$
    Compute $\boldsymbol{i}_{\alpha\beta}(k+1)(j)$
    $G(j) \leftarrow (i_{\alpha ref} - i_\alpha)^2 + (i_{\beta ref} - i_\beta)^2$
    if $G(j) < G_{op}$ **then**
      $G_{op} \leftarrow G(j)$
      $j_{op} \leftarrow j$
    end if
  end for
    return $u(j_{op})$

III. TWO NEW PROPOSALS FOR FINITE CONTROL SET MODEL PREDICTIVE CONTROL

In this paper, two new FCS-MPC techniques are proposed that also use a reduced set of admissible voltage vectors. The specific choice of voltage vectors depends on the sector in which the motor is operating and on the direction of rotation. These sectors, in turn, are related to the electrical position of the rotor, as shown in (4). Two of the admissible voltage vectors are selected such that they are adjacent to the sector where the motor is currently located, and the other one is a zero vector, in a similar way to Space Vector Modulation, giving a total of 3 admissible voltage vectors for each of the 6 sectors, as shown below.

$$\mathcal{U} = \begin{cases} V_0, V_2, V_3 & for \quad -\frac{\pi}{6} \leq \theta_e \leq \frac{\pi}{6} \\ V_0, V_3, V_4 & for \quad \frac{\pi}{6} \leq \theta_e \leq \frac{3\pi}{6} \\ V_0, V_4, V_5 & for \quad \frac{3\pi}{6} \leq \theta_e \leq \frac{5\pi}{6} \\ V_0, V_5, V_6 & for \quad \frac{5\pi}{6} \leq \theta_e \leq \frac{7\pi}{6} \\ V_0, V_6, V_1 & for \quad \frac{7\pi}{6} \leq \theta_e \leq \frac{9\pi}{6} \\ V_0, V_1, V_2 & for \quad \frac{9\pi}{6} \leq \theta_e \leq \frac{11\pi}{6} \end{cases} \quad (4)$$

where θ_e is the rotor electrical position. Using this set of vectors reduces not only the number of switching actions but also the number of times the algorithm has to be executed.

A. FCS-M[1]PC

The first technique, named FCS-M[1]PC, is based on finding a voltage vector formed by the combination of one of the two adjacent vectors to the current sector with the zero vector, such that their combination minimizes the cost function. This optimal combination is then applied for the entire sampling interval as will be explained below.

To avoid the calculation of the switching time between the non-zero vector and the zero vector during each sampling time, that is used in most techniques proposed in the

literature, and that increases de complexity and execution time of the algorithm, our approach maintains the simplicity of the original technique FCS-MPC by dividing the sampling time into a fixed number of equal sub-sampling times and evaluating for each one these sub-intervals the effect in the cost function of using one of the adjacent vectors, that correspond to the sector in which the rotor is located, or the zero vector.

Fig. 1 illustrates the operation of the algorithm, where in this example the sampling interval is divided into four equal sub-intervals. It is clear from the figure, that for the first three sub-intervals V_1 is the best choice, and that in the fourth, the vector V_0 should be applied.

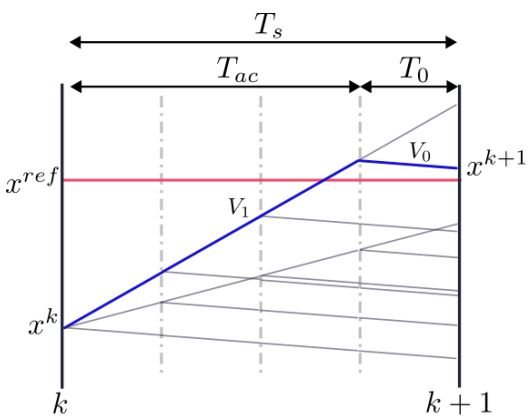


FIGURE 1. Calculation of the optimal vector in FCS-M^{1}PC with the sampling time divided into four equal sub-intervals.

The predictive model is obtained by discretizing the electrical part of the motor's mathematical model in a fashion similar to Eq.(2) of the classical FCS-MPC approach, but with some minor modifications, as shown below.

$$i_\alpha(k+1) = i_\alpha(k) + \frac{1}{L_s}\Big(T_{ac}u_\alpha(k)\Big) - \frac{T_s}{L_s}\Big(e_\alpha(k) + Ri_\alpha(k)\Big) \tag{5a}$$

$$i_\beta(k+1) = i_\beta(k) + \frac{1}{L_s}\Big(T_{ac}u_\beta(k)\Big) - \frac{T_s}{L_s}\Big(e_\beta(k) + Ri_\beta(k)\Big) \tag{5b}$$

where T_{ac} is the time when the non-zero vector is applied given by

$$T_{ac} = \frac{jT_s}{M}; j \in \{0, \dots \rho - 1\},$$

with ρ the number of selected partitions for the sampling time T_s. Note that Eq (5) has to be solved ρ times, for for $j \in \{0, \dots \rho - 1\}$ and that the constraint $T_{ac} + T_0 = T_s$ must be satisfied, with T_0 the time during which the zero vector is active. The cost function is in Eq. (3).

With this technique, there is flexibility in determining the number of time divisions ρ, which defines how many times the algorithm must be executed. For example, by dividing the sampling time into seven equal parts and using the reduced set of admissible voltage vectors per sector (4), with two active voltage vectors and one zero vector, a total of 15 different voltage vectors are obtained. One corresponds to the zero vector applied throughout the entire sampling interval, seven correspond to one of the active vectors applied for different fractions of the sampling time, and the remaining seven correspond to the other active vector applied in the same manner. Fig. 2 illustrates the distribution of the generated vectors in the $\alpha\beta$ plane, where each point in the figure represents one of these seven vectors mentioned above.

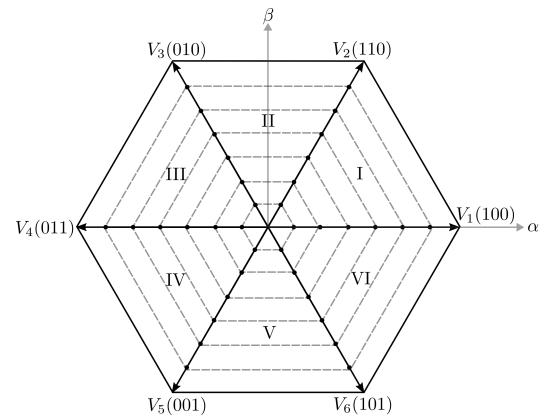


FIGURE 2. FCS-M^{1}PC with 7 divisions. Dots represent all possible choices of active vector for all rotor positions.

In this seven-part example, the algorithm simply evaluates each of the 15 generated vectors, selects the one that minimizes the cost function, and applies it in the next sampling instant. The pseudocode of the algorithm is presented below.

Algorithm 2 FCS-M^{1}PC

Require: $i(k), u(k)$
  $G_{op} \leftarrow G_{max}$
  $j \leftarrow 1$
  for $j \leftarrow 1, 15$ **do**
  Select the j combination of vectors and sampling time fractions $(\boldsymbol{u}_{\alpha\beta}, T_{act})(j) = (\boldsymbol{u}_{1\alpha\beta}, T_1), ..., (\boldsymbol{u}_{1\alpha\beta}, T_7)$, $(\boldsymbol{u}_{2\alpha\beta}, T_1), ..., (\boldsymbol{u}_{2\alpha\beta}, T_7), (\boldsymbol{u}_{0\alpha\beta}, T_s)$
    Compute $\boldsymbol{i}_{\alpha\beta}(k+1)(j)$
    $G(j) \leftarrow (i_{\alpha ref} - i_\alpha(k+1))^2 + (i_{\beta ref} - i_\beta(k+1))^2$
    if $G(j) < G_{op}$ **then**
      $G_{op} \leftarrow G(j)$
      $j_{op} \leftarrow j$
      $(\boldsymbol{u}_{op\alpha\beta}, T_{opact}) \leftarrow (\boldsymbol{u}_{\alpha\beta}, T_{act})(j_{op})$
    end if
  end for
    return Modulates combination j_{op}.

A larger number of sub-sampling time divisions results in a lower tracking error; however, it also significantly increases

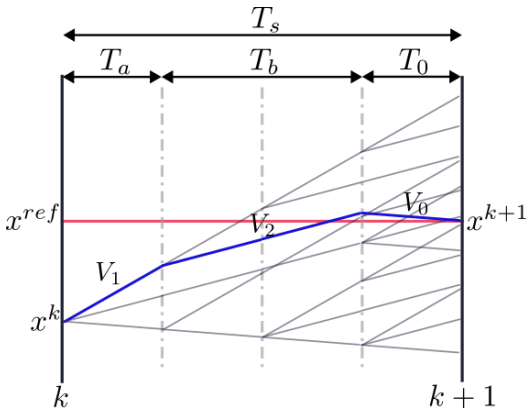


FIGURE 3. Calculation of the optimal vector in FCS-M^{2}PC with the sampling time divided into four equal subsamplings.

the number of algorithm executions and, consequently, its execution time. Therefore, it is essential to strike a balance between the number of sampling time divisions and the algorithm complexity, considering the hardware on which it will be implemented.

It could be argued that a similar result could be obtained in the example of Fig. 1 by choosing a sampling time of one quarter of our sampling time. This could be correct for large sampling times, however when running at very small sampling times, the time overhead incurred when handling interruptions starts to have a direct impact on the number of possible calculations before the next interruption occurs. This overhead takes a larger proportion of the sampling time as it is reduced, limiting the feasible calculations[2].

B. FCS-M^{2}PC

The other algorithm, FCS-M^{2}PC, can be seen as an extension of the previous one. In this case, the algorithm selects a combination of the two voltage vectors adjacent to the sector position and the zero vector that minimizes the cost function that results from applying this combination in the next sampling period, and determines the time each vector should remain active. In a similar fashion to FCS-M^{1}PC, this is achieved dividing the sampling time into equal sub-interval and by testing a finite number of combinations of the two active vectors, varying the fraction of time each one remains active, and completing the sampling time T_s by applying the zero vector. Fig. 3 provides a simplified representation of this algorithm operation while tracking an arbitrary reference.

The same cost function as in the previous technique is used, while the predictive model is modified as follows.

$$i_\alpha(k+1) = i_\alpha(k) + \frac{1}{L_s}\Big(T_a u_{a\alpha}(k) + T_b u_{b\alpha}(k)\Big) - \frac{T_s}{L_s}\Big(e_\alpha(k) + R i_\alpha(k)\Big) \tag{6a}$$

$$i_\beta(k+1) = i_\beta(k) + \frac{1}{L_s}\Big(T_a u_{a\beta}(k) + T_b u_{b\beta}(k)\Big) - \frac{T_s}{L_s}\Big(e_\beta(k) + R i_\beta(k)\Big) \tag{6b}$$

where the two adjacent vectors in a sector are denoted as u_a and u_b, meaning that the terms $u_{a\alpha}, u_{a\beta}, u_{b\alpha}, u_{b\beta}$ represent the voltage components in the $\alpha\beta$ reference frame corresponding to each adjacent vector. The times T_a and T_b indicate the duration for which each vector, u_a and u_b remains active, respectively. These times satisfy the equation $T_a + T_b + T_0 = T_s$, where T_0 represents the time that the zero vector remains active, that it is also used to measure the currents.

The number of times Eq. (6) must be calculated is determined by the total number of combinations without repetition of the admissible voltage vectors (n) and the divisions of the sampling time (ρ). This can be calculated using the following expression:

$$C^R(n,\rho) = \frac{(n+\rho-1)!}{\rho!(n-1)} \tag{7}$$

In this case, if the 3 admissible voltage vectors presented in (4) are used, along with 4 divisions of the sampling time, this results in a total of 15 combinations without repetition to test for each sampling interval. Fig. 4 shows the vectors generated by these combinations, represented as points in the $\alpha\beta$ plane.

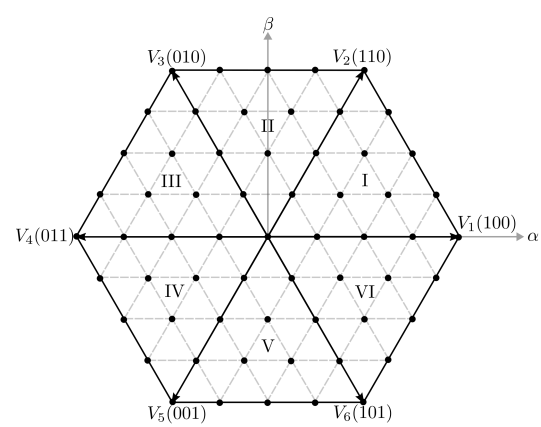


FIGURE 4. FCS-M^{2}PC with 4 divisions. Dots represent all possible choices of active vector for all rotor positions.

Again, increasing the number of sub-intervals ρ can improve controller's performance, however, significantly increasing the computational load, within hardware capacity. The following pseudocode illustrates the operation of the proposed algorithm.

[2]At the limit, the overhead in a too small sampling time does not give time for any calculation, i.e., there is a maximum sampling time a given hardware can sustain, otherwise overrun happens.

Algorithm 3 FCS-M^{2}PC

Require: $i(k), u(k)$
  $G_{op} \leftarrow G_{max}$
  $j \leftarrow 1$
  for $j \leftarrow 1, C^R(3, \rho)$ **do**
  Select the j combination of vectors $\boldsymbol{u}_{1\alpha\beta}(k), ..., \boldsymbol{u}_{\rho\alpha\beta}(k)$
    Compute $\boldsymbol{i}_{\alpha\beta}(k+1)(j)$
    $G(j) \leftarrow (i_{\alpha ref} - i_\alpha)^2 + (i_{\beta ref} - i_\beta)^2$
    if $G(j) < G_{op}$ **then**
      $G_{op} \leftarrow G(j)$
      $j_{op} \leftarrow j$
    end if
  end for
    return Modulate the j_{op} combination of VV.

The finite combination of vectors can be implemented using a lookup table containing the activation times for each of the adjacent vectors. The algorithm uses this table to compute the resulting voltage vector for a given combination in the $\alpha\beta$ reference frame, predicts the currents, evaluates the cost function, and determines which voltage vector combination minimizes it. This process is repeated for each vector combination in the lookup table. The combination that minimizes the cost function is then modulated and applied to the inverter.

The proposed FCS-M^{2}PC technique requires applying one of the three admissible voltage vectors for a period shorter than the sampling time or even combining all of them within the same time interval. This requires some form of modulation. In this case, the modulation of the vectors is performed similarly to space vector modulation (SVM).

The previous algorithms simply determine the time during which each vector should remain active, using a reduced set of voltage vectors that change per sector. There are 6 sectors, which vary depending on the rotor's position. The sectors and the corresponding admissible voltages for each sector were selected in such a way that they align with those used in SVM.

This modulation technique usually requires calculating the times during which the adjacent vectors must remain active in a given sector. However, by using this algorithm, it is no longer necessary to calculate these times. Once the optimal vector combination is known, it becomes relatively easy to determine the durations for which the adjacent and zero vectors should remain active. The resulting vector V_s is calculated as follows:

$$V_s = \frac{1}{T_s}(V_a T_a + V_b T_b + V_0 T_0). \tag{8}$$

With the knowledge of the sector and the times the adjacent vectors should remain active, it is possible to determine how long each phase should stay active in the original reference frame to generate the desired vector. This is achieved using Table. II.

TABLE II. Calculation of Phase Time in SVM

Sector	T_A	T_B	T_C
1	$T_0 + T_a$	$T_0 + T_a + T_b$	T_0
2	T_0	$T_0 + T_a + T_b$	$T_0 + T_b$
3	T_0	$T_0 + T_a$	$T_0 + T_a + T_b$
4	$T_0 + T_b$	T_0	$T_0 + T_a + T_b$
5	$T_0 + T_a + T_b$	T_0	$T_0 + T_a$
6	$T_0 + T_a + T_b$	$T_0 + T_b$	T_0

IV. SIMULATIONS RESULTS

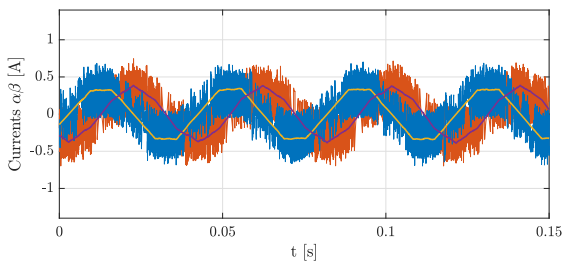
To demonstrate the effectiveness of the two proposed algorithms, FCS-M^{1}PC and FCS-M^{2}PC, numerical validation are presented, whose results are compared to the classical FCS-MPC technique. The simulations were performed in Simulink MATLAB using the parameters obtained from the datasheet of the BLDC motor BLY-344S-240V-3000 from Anaheim Automation, as shown in Table. III.

TABLE III. Parameters of BLDC motor BLY-344S-240V-3000

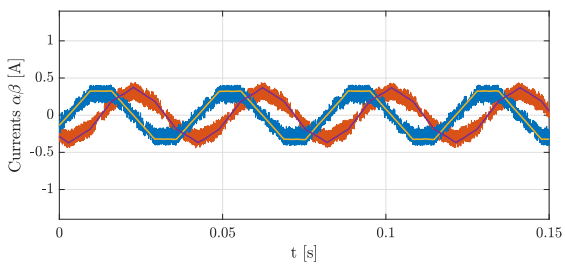
Parameter	Value
Rated Voltage	240 [V]
Rated Torque	2.1 [N m]
Rated Power	600 [W]
Resistance	1.2 [Ω]
Inductance	0.00205 [mH]
Electric constant	0.40355 [V/rad/s]
Mechanical constant	0.65997 [Nm/A]
Inertia	0.00027948 [kgm^2]
Viscous friction coefficient	0.0006738 [N m s]

To verify the performance of the algorithms, several simulations were carried out under different operating conditions. First, two sets of simulations were performed. In the first one, the motor was assumed to have a trapezoidal BEMF; therefore, trapezoidal current trajectories were used to reduce current and torque ripple. In the second set, the motor was assumed to have a sinusoidal BEMF, and accordingly, the current trajectories were also sinusoidal. It was also assumed that the inverter is ideal and that its branches are controlled by pulse-width modulation generated through space vector modulation (SVM). The sampling time used for current control was $T_s = 3 \times 10^{-5} s$ for all techniques. Since the objective is solely to compare current behavior, the motor was simulated operating at a constant speed of 40 rad/s with

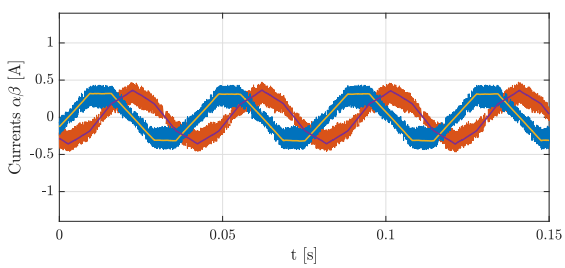
a small, constant load. Subsequently, the algorithms were tested at different speeds to evaluate their performance, and finally, another series of simulations was carried out varying the number of divisions in the algorithms to assess their effect. For simplicity and to avoid overloading the document with figures, only the results of selected experiments are presented in graphical form. The remaining results are summarized in tables showing the corresponding error values.



(a) FCS-MPC



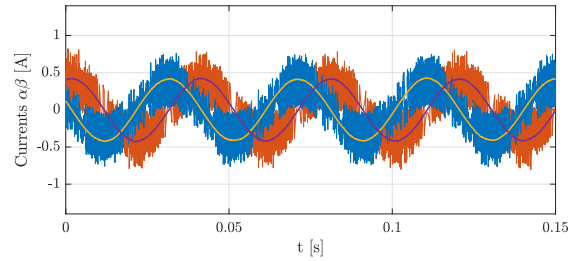
(b) FCS-M^{1}PC



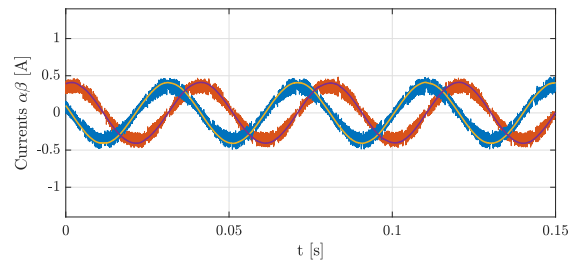
(c) FCS-M^{2}PC

FIGURE 5. Simulation results of measured and reference currents for a motor with Trapezoidal BEMF.

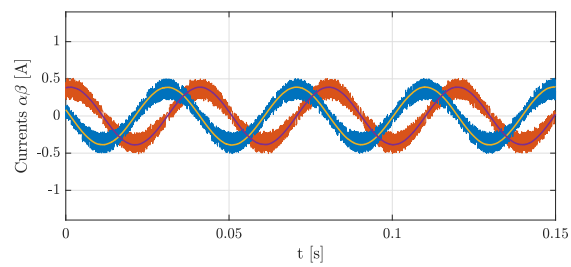
Figures 5 and 6 illustrate the behavior of the currents, together with their reference values in the $\alpha\beta$ reference frame, for the FCS-MPC, FCS-M^{1}PC, and FCS-M^{2}PC techniques. One set of results corresponds to the motor with trapezoidal BEMF (Fig. 5), while the other corresponds to the motor with sinusoidal BEMF (Fig. 6). Meanwhile, Tables IV and V present the root mean square error (RMSE) between the reference and measured currents for both sets of simulations. In addition, Table. VI shows the normalized mean squared error (NMSE) of the current, presented in (9), for the different algorithms at various speeds, allowing a performance



(a) FCS-MPC



(b) FCS-M^{1}PC



(c) FCS-M^{2}PC

FIGURE 6. Simulation results of measured and reference currents for a motor with Sinusoidal BEMF at 40 rad/seg.

comparison under different operating conditions.

$$\text{NMSE} = \frac{\sum_{i=1}^{n}(i_\rho - i_{\rho ref})^2}{\sum_{i=1}^{n}(i_\rho)^2};\ \rho = \alpha, \beta \tag{9}$$

Finally, Tables VII and VIII summarize the RMSE of the currents obtained with the FCS-M^{1}PC and FCS-M^{2}PC algorithms using different numbers of divisions.

TABLE IV. Simulation RMSE values for Trapezoidal BEMF motor at 40 rad/seg.

Technique	RMSE i_α	RMSE i_β
FCS-MPC	0.1723	0.1678
FCS-M^{1}PC	0.0548	0.0559
FCS-M^{2}PC	0.0598	0.0608

From Tables IV and V, it is evident that the two proposed techniques exhibit a lower current trajectory tracking error

TABLE V. Simulation RMSE values for Sinusoidal BEMF motor at 40 rad/seg.

Technique	RMSE i_α	RMSE i_β
FCS-MPC	0.1661	0.1671
FCS-M^{1}PC	0.0529	0.0518
FCS-M^{2}PC	0.0567	0.0574

TABLE VI. Simulation NMSE values for Sinusoidal BEMF motor at various speeds (rad/seg).

Technique	Speed	NMSE i_α	NMSE i_β
FCS-MPC	40	0.3212	0.3200
FCS-M^{1}PC	40	0.0325	0.0335
FCS-M^{2}PC	40	0.0430	0.0446
FCS-MPC	80	0.2319	0.2284
FCS-M^{1}PC	80	0.0534	0.0549
FCS-M^{2}PC	80	0.0417	0.0400
FCS-MPC	100	0.1902	0.2041
FCS-M^{1}PC	100	0.0611	0.0599
FCS-M^{2}PC	100	0.0373	0.0384

compared to the classical FCS-MPC technique, that would need to operate at a significantly lower sampling time to achieve performance similar to the proposed techniques. It can also be observed in Table. VI that the NMSE of the FCS-M^{2}PC technique decreases as the operating speed increases, while the FCS-M^{1}PC technique only shows a lower NMSE than FCS-M^{2}PC at lower speeds. Furthermore, from Tables VII and VIII it can be seen that increasing the number of divisions in the FCS-M^{1}PC technique does not produce a significant effect on the RMSE. In contrast, the FCS-M^{2}PC technique exhibits a noticeable reduction in the RMSE of the currents as the number of divisions increases.

The lower the inductance of the motor, the higher the execution frequency required for the algorithms, making the implementation of the classical FCS-MPC more challenging

TABLE VII. Simulation RMSE results of FCS-M^{1}PC for different T_s divisions at 40 rad/seg.

Divisions of T_s	# of tests	RMSE i_α	RMSE i_β
4	9	0.0611	0.0628
7	15	0.0569	0.0566
10	21	0.0634	0.0615

TABLE VIII. Simulation RMSE results of FCS-M^{2}PC for different T_s divisions at 40 rad/seg.

Divisions of T_s	# of tests	RMSE i_α	RMSE i_β
3	10	0.0699	0.0682
4	15	0.0573	0.0577
5	21	0.0491	0.0502

in motors with low inductance. The proposed techniques mitigate this issue by reducing the need for extremely small sampling times, albeit at the cost of a slightly increased computational load.

V. EXPERIMENTAL RESULTS

To validate the proposed algorithms, the experimental platform shown in Fig. 7 was built. It consists of a BLDC motor, an encoder, a torque sensor, and a DC motor to generate load torque when needed.

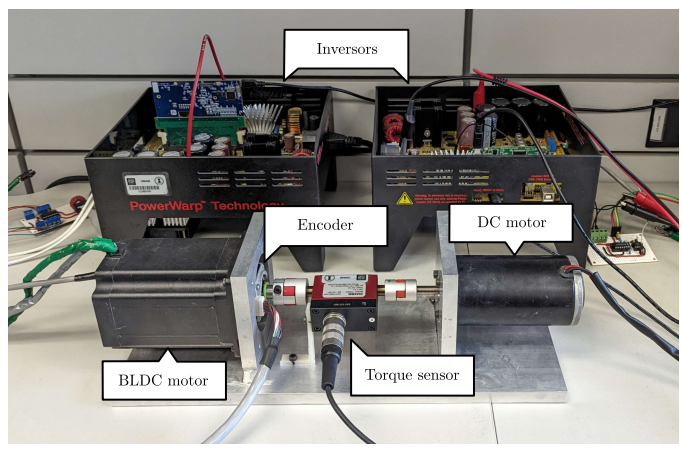


FIGURE 7. Experimental test bench.

The BLDC motor used was also the BLY344S-240V-3000 model from Anaheim Automation. The absolute capacitive encoder AMT333S-V from CUI Devices was coupled to its shaft. This encoder offers a configurable resolution ranging from 48 to 4096 pulses per revolution. In this case, it was set to a resolution of 2048 PPR and was used to determine the rotor position and calculate its speed.

The inverter employed was the high-voltage digital motor control and power factor correction kit (TMDSHVMTRPFCKIT) from Texas Instruments (TI). This board includes an AC rectifier stage to generate the DC bus voltage and a three-phase inverter stage designed to control motors with voltages up to 450 V DC.

The current measurement is performed on the low side of each inverter leg, that is, between the emitter of the lower transistor and the negative DC bus, using shunt resistors. However, this configuration presents a limitation: the current

can only be measured in specific transistor states, specifically when the lower transistor of a leg is activated.

To address this issue, a period of time in which a null vector is applied was added to the proposed current control algorithms in some cases[3], ensuring that the current measurement is performed correctly during each sampling interval.

The control card used was based on the F28379D Delfino microcontroller, also from TI. This is a 32-bit floating-point microcontroller with dual CPUs, optimized for real-time control applications. The microcontroller was programmed using the Simulink-Matlab code generation tool, which facilitated the implementation of the new control algorithms.

Following a similar approach to the simulations, the experimental tests compared the classical FCS-MPC technique with the two proposed variants for current control, FCS-M^{1}PC and FCS-M^{2}PC. One experiment employed trapezoidal current trajectories to drive the motor, while another used sinusoidal current trajectories. The sampling time used for all techniques was $T_s = 3\times10^{-5}s$. The experimental procedure was designed to replicate the simulation conditions, including a baseline comparison at 40 rad/s with a small, constant load torque applied by the DC motor, evaluations at different operating speeds, and a comparison of the algorithms using different numbers of time divisions. As in the simulations, only representative experimental waveforms are provided, while the remaining results are summarized in tables showing the corresponding error values.

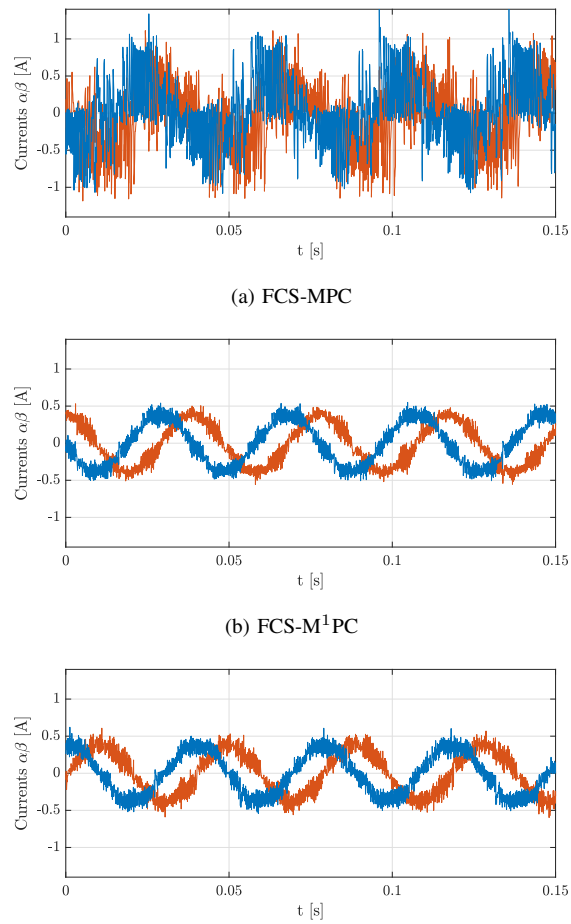
Figures 8 and 9 illustrate the measured current behavior in the $\alpha\beta$ reference frame for the three control techniques. One set of results corresponds to experiments in which the motor was driven with trapezoidal current trajectories, while the other corresponds to experiments using sinusoidal trajectories. Tables IX and X present the RMSE of the measured currents for both sets of experiments. In addition, Table. XI summarizes the NMSE of the currents obtained with the different algorithms at various speeds. Finally, Tables XII and XIII show the RMSE of the currents for the FCS-M^{1}PC and FCS-M^{2}PC techniques when varying the number of divisions.

TABLE IX. Experimental RMSE values using trapezoidal current trajectories at 40 rad/seg.

Technique	RMSE i_α	RMSE i_β
FCS-MPC	0.3363	0.3430
FCS-M^{1}PC	0.0691	0.0710
FCS-M^{2}PC	0.0819	0.0828

In general, as usual, the experimental errors are noticeably higher than those observed in the simulations. This discrepancy can be attributed to several factors, including higher than expected noise in the current measurements, parameter uncertainties, and variations in the inductance.

[3]For example, when in FCS-MPC or FCS-M^{2}PC the zero voltage is not applied



(a) FCS-MPC

(b) FCS-M^{1}PC

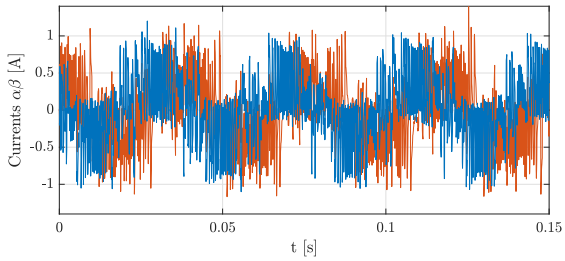
(c) FCS-M^{2}PC

FIGURE 8. Experimental results of measured currents using trapezoidal current trajectories at 40 rad/seg.

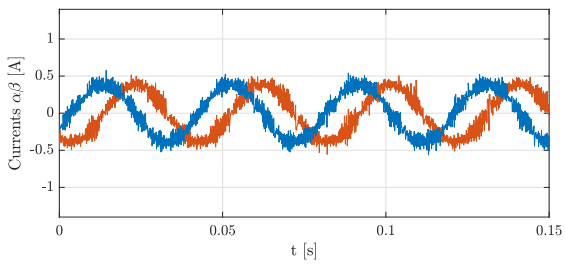
TABLE X. Experimental RMSE values using sinusoidal current trajectories at 40 rad/seg.

Technique	RMSE i_α	RMSE i_β
FCS-MPC	0.3517	0.3706
FCS-M^{1}PC	0.0703	0.0727
FCS-M^{2}PC	0.0772	0.0786

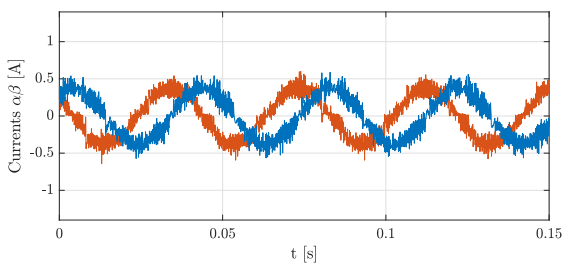
Despite these factors, as shown in Tables IX and X, the proposed techniques still achieve significantly lower errors compared to the classical FCS-MPC method. Table. XI indicates that between the two proposed approaches, FCS-M^{2}PC attains the lowest error at higher speeds, whereas FCS-M^{1}PC performs better at lower speeds. In contrast with the simulation results, error size increases with speed.



(a) FCS-MPC



(b) FCS-M^{1}PC



(c) FCS-M^{2}PC

FIGURE 9. Experimental results of measured currents using sinusoidal current trajectories at 40 rad/seg.

Furthermore, from Tables XII and XIII, the behavior of both algorithms with different numbers of divisions is consistent with that observed in the simulations: the FCS-M^{2}PC technique shows a noticeable reduction in the RMSE of the currents as the number of divisions increases, while FCS-M^{1}PC exhibits almost no change.

VI. CONCLUSION

Two new techniques for FCS-MPC of BLDC motors were presented that by using a reduced set of possible applied voltages and dividing the sampling time into a fixed number of sub-intervals allow to obtain important current ripple reductions, when compared with the traditional FCS-MPC technique. The proposals slightly increase the computational load required to calculate the optimal choice of voltage vectors that minimizes a predefined cost function, however they still keep the simplicity of the classical FCS-MPC. Simulation and experimental results under similar conditions showed a reduction between 75% to 80% in the RMSE current tracking error, compared to the classical FCS-MPC.

TABLE XI. Experimental NMSE values for sinusoidal BEMF motor at various speeds (rad/seg).

Technique	Speed	NMSE i_α	NMSE i_β
FCS-MPC	40	0.6438	0.6564
FCS-M^{1}PC	40	0.0622	0.0661
FCS-M^{2}PC	40	0.0753	0.0787
FCS-MPC	80	0.5073	0.5178
FCS-M^{1}PC	80	0.1275	0.1338
FCS-M^{2}PC	80	0.0907	0.0972
FCS-MPC	100	0.4882	0.5092
FCS-M^{1}PC	100	0.1689	0.1750
FCS-M^{2}PC	100	0.1135	0.1155

TABLE XII. Experimental RMSE results of FCS-M^{1}PC for different T_s divisions at 40 rad/seg.

Divisions of T_s	# of tests	RMSE i_α	RMSE i_β
4	9	0.0761	0.0792
7	15	0.0710	0.0731
10	21	0.0706	0.0723

A more extensive testing of the proposals is still required in combination with velocity tracking schemes with load uncertainty.

ACKNOWLEDGMENT

This work was supported in part by the UNAM-PAPIIT grant IT100623 and SECIHTI Grant 123456. Authors thank Enrique Gómez Rosas for his help with the experimental platform.

REFERENCES

[1] M. Ehsani, Y. Gao, S. Longo and K. Ebrahim, *Modern electric, hybrid electric, and fuel cell, vehicles* CRC press, 2018.
[2] T.J. Zarma, A.A. Galadima and M.A. Aminu, *Review of Motors for Electrical Vehicles*, Journal of Scientific Research and Reports (2019), 1–6.

TABLE XIII. Experimental RMSE results of FCS-M^{2}PC for different T_s divisions at 40 rad/seg.

Divisions of T_s	# of tests	RMSE i_α	RMSE i_β
3	10	0.0943	0.0977
4	15	0.0780	0.0791
5	21	0.0665	0.0715

[3] Tabassum, A., Cho, H.M., Mahmud, M.I., *Essential Features and Torque Minimization Techniques for Brushless Direct Current Motor Controllers in Electric Vehicles*, Energies **17** (2024), no. 18, 4562.

[4] Microchip Technology Inc., *Comparison between FOC and six-step control*, Technical documentation, Microchip (2025).

[5] Le-Huy, H., *Minimization of Torque Ripple in Brushless DC Motor Drives*, Proceedings of the IEEE IAS Annual Meeting, 1986.

[6] Chen, Y., Zhou, Y., Zhang, J., *Direct torque control of brushless DC motor based on improved predictive control*, Journal of Electrical Engineering **64** (2013), no. 3, 151–157.

[7] Wu, Z., Zhao, S., Xu, X., Wang, J., *Direct Torque Control Strategy for Brushless DC Motor Based on the Current Sampled in DC Link*, IEEE Access **6** (2018), 47687–47696.

[8] Arias, A., Watanabe, T., Kawasaki, T., *Hysteresis current control of a BLDC motor using an adaptive controller*, IEEE IECON 2001, vol. 2, 2001, pp. 1215–1220.

[9] Patel, M., Patel, R., *Hysteresis current control of BLDC motor drive with improved torque performance*, International Journal of Power Electronics and Drive Systems **11** (2020), no. 4, 2095–2103.

[10] M. S. Trivedy adn R. K. Keshri. *Evaluation of predictive current control techniques for PM BLDC motor in stationary plane* IEEE Access (2020), 8:46217-46228, DOI 10.1109/ACCESS.2020.2978695.

[11] M. F. Elmorshedy, M.F. Xu, F. F. El-Sousy, M. R. Islam and A. A. Ahmed *Recent achievements in model predictive control techniques for industrial motor: A comprehensive state-of-the-art* IEEE Access (2021), 9:58170-58191, DOI 10.1109/ACCESS.2021.3073020.

[12] de Castro, A.G., de Andrade Pereira, W.C., de Oliveira, C.M., de Almeida, T.E., Guazzelli, P.R., de Almeida Monteiro, J.R., de Oliveira Junior, A.A., *Finite control-set predictive power control of BLDC drive for torque ripple reduction*, IEEE Latin America Transactions **16** (2018), 1128–1135.

[13] Šlapák, J., et al., *Finite Control Set Model Predictive Speed Control of a DC Motor*, Mathematical Problems in Engineering **2016** (2016), Article ID 9571972.

[14] J. Rodriguez, M. P. Kazmierkowski, J. R. Espinoza, P. Zanchetta, H. Abu-Rub, H. A. Young and C. A. Rojas *State of the art of finite control set model predictive control in power electronics* IEEE Transactions on Industrial Informatics (2013), 9(2):1003-1016, DOI 10.1109/TII.2012.2221469.

[15] Xia, C., et al., *Model predictive current control of PMSM using dual-vector control with duty cycle optimization*, IEEE Transactions on Industrial Electronics **67** (2020), no. 4, 2683–2693.

[16] Wen, X., et al., *Model predictive torque control with duty ratio optimization for PMSM based on multiple virtual voltage vectors*, IEEE Transactions on Industrial Electronics **67** (2020), no. 12, 10237–10247.

[17] de Castro, A.G., et al., *Model predictive control strategies for torque ripple minimization of PMSM drives*, IEEE Transactions on Industrial Applications **54** (2018), no. 6, 6215–6224.

[18] Li, K., Liu, Y., Wei, B., Wang, P., *Multi-Virtual-Vector Model Predictive Current Control for Dual Three-Phase PMSM*, Energies **14** (2021), no. 21, 7292.

[19] Zhang, X., Sun, Y., Chen, J., *Low-Complexity Model Predictive Control for Series-Winding PMSM with Extended Voltage Vectors*, Electronics **14** (2022), no. 1, 127.

[20] Sheng, K., et al., *Two-Vector FCS-MPC for Permanent-Magnet Synchronous Motors Based on Duty Ratio Optimization*, Mathematical Problems in Engineering **2018** (2018), Article ID 9061979.

[21] Wang, R., et al., *An Improved Double-Vector MPC Strategy for PMSM Considering DC Voltage Pulsation*, IET Electric Power Applications **18** (2024), e12460.

[22] Zhao, L., et al., *Enhanced Dual-Vector Model Predictive Control for PMSM Drives Using an Optimal Vector Selection Principle*, Energies **16** (2022), no. 22, 7482.

[23] J.N. Chiasson, *Modeling and high performance control of electric machines*, John Wiley & Sons, 2005.

AXEL CORONADO-ANDRADEreceived the B.S. degree in Mechatronics Engineering from the Instituto Tecnológico y de Estudios Superiores de Monterrey. His M.Sc. and Ph.D. degrees in Electrical Engineering (Automatic Control) are from the Universidad Nacional Autónoma de México (UNAM). His research interests include estimation and control of electric machines.

LUIS ALVAREZ-ICAZA(Member, IEEE) received a B.Sc. in Mechanical Engineering, an M.Sc. in Control Engineering from the National Autonomous University of Mexico (UNAM), and a Ph.D. in Mechanical Engineering from the University of California at Berkeley. He is currently a researcher for the National Autonomous University of Mexico. In addition, he is an Emeritus National Researcher (granted by the National Research System - SNII by its acronym in Spanish). His research interests include control theory and modeling, estimation and control of thermo-chemical processes and electromechanical systems, and complex system modeling.